GATE ME 90 Days Challenge

Day 85/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	Full-Length Mock 6	12 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed full-syllabus warm-up for the sixth full-length mock: antonyms, a clock that gains time, the inverse of a 2 × 2 matrix, Euler's method, uniformly accelerated rotation, strain energy in stepped bars, shear stress in rectangular beams, instantaneous centres, springs in series, the Soderberg criterion, laminar pipe flow, pin-fin efficiency, polytropic work, cold working and $\bar{x}$ control charts. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the word that is most nearly OPPOSITE in meaning to TACITURN.

(A) Reserved
(B) Talkative
(C) Silent
(D) Sullen

Q2 MCQ

A wall clock gains 5 minutes every hour. It is set to the correct time at 8:00 a.m. When the correct time is 8:00 p.m. the same day, the clock shows

(A) 8:55 p.m.
(B) 9:00 p.m.
(C) 7:00 p.m.
(D) 9:05 p.m.

Q3 MCQ

The inverse of the matrix with rows (3, 1) and (5, 2) has rows

(A) (2, −1) and (−5, 3)
(B) (2, 1) and (5, 3)
(C) (−2, 1) and (5, −3)
(D) (3, −1) and (−5, 2)

Q4 NAT

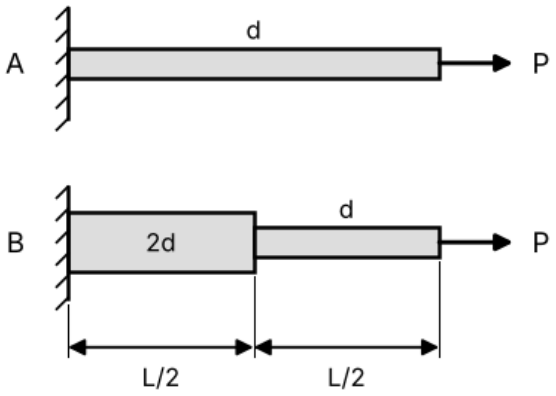
The equation $dy/dx = 2x - y$ with $y(0) = 1$ is solved by Euler's (explicit) method with a step $h = 0.1$. The value of y at $x = 0.2$ (round off to two decimal places) is _____

Q5 NAT

A wheel starts from rest and accelerates uniformly to 600 rpm in 10 s. The number of revolutions it makes in this time is _____

Q6 MCQ

Bars A and B, shown, are of the same material and length L. Bar A has a uniform diameter d. Bar B has diameter d over half its length and 2d over the other half. Both carry axial loads that produce the same maximum stress σ. The ratio of the strain energy stored in B to that stored in A is



(A) 1/2
(B) 5/8
(C) 5/16
(D) 1

Q7 NAT

A beam of rectangular section 120 mm wide and 240 mm deep carries a vertical shear force of 72 kN. The transverse shear stress at a level 60 mm above the neutral axis (in MPa, round off to two decimal places) is _____

Q8 MCQ

For a four-bar chain (including the fixed link), the total number of instantaneous centres and the number of them that lie at the pin joints are

(A) 4 and 4
(B) 6 and 2
(C) 8 and 4
(D) 6 and 4

Q9 NAT

A 5 kg mass hangs from two springs of stiffness 2 kN/m and 3 kN/m connected in series. The natural frequency of the system (in rad/s, round off to two decimal places) is _____

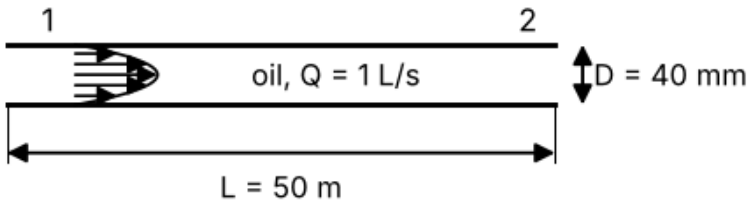
Q10 MCQ

A machine part made of a steel with yield strength 300 MPa and corrected endurance limit 200 MPa carries a fluctuating stress with a mean of 100 MPa and an amplitude of 50 MPa. The factor of safety by the Soderberg criterion is about

(A) 1.71
(B) 2.40
(C) 1.20
(D) 3.00

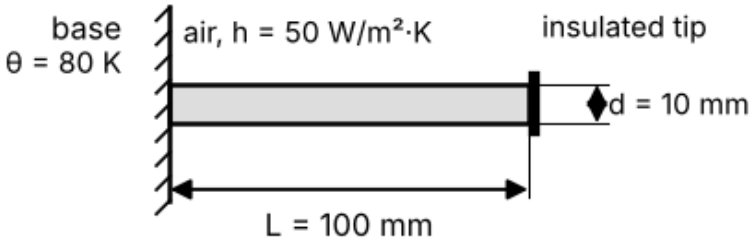
Q11 NAT

Oil (density 900 kg/m³, dynamic viscosity 0.1 Pa·s) flows at 1 litre per second through a horizontal pipe of 40 mm diameter and 50 m length. Assuming fully developed laminar flow, the pressure drop over the length (in kPa, round off to one decimal place) is _____



Q12 MCQ

An aluminium pin fin (k = 200 W/m·K) of 10 mm diameter and 100 mm length has an insulated tip, and h = 50 W/m²·K. The base is 80 K hotter than the surrounding air. The fin efficiency and the heat lost by the fin are about



(A) 0.76 and 12.6 W
(B) 0.96 and 9.6 W
(C) 0.76 and 9.6 W
(D) 0.62 and 7.8 W

Q13 NAT

Air is compressed in a closed cylinder from 100 kPa and 0.3 m³ to 0.1 m³ following $pV^{1.3}$ = constant. The work done on the air (in kJ, round off to one decimal place) is _____

Q14 MCQ

When a metal is cold worked (plastically deformed below its recrystallisation temperature),

(A) its strength and hardness rise while its ductility falls
(B) its strength and ductility both rise
(C) its hardness falls and its ductility rises
(D) its electrical conductivity rises sharply

Q15 NAT

Samples of 5 parts each are taken from a process. The grand mean of the sample means is 50.00 mm and the average sample range is 4.00 mm. Taking A_2 = 0.577 for n = 5, the upper control limit of the $\bar{x}$ chart (in mm, round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(B) Talkative
Q2	MCQ	(B) 9:00 p.m.
Q3	MCQ	(A) (2, −1) and (−5, 3)
Q4	NAT	0.83
Q5	NAT	50
Q6	MCQ	(B) 5/8
Q7	NAT	2.81 (accept 2.79 to 2.83)
Q8	MCQ	(D) 6 and 4
Q9	NAT	15.49 (accept 15.40 to 15.58)
Q10	MCQ	(A) 1.71
Q11	NAT	79.6 (accept 79.2 to 80.0)
Q12	MCQ	(C) 0.76 and 9.6 W
Q13	NAT	39.0 (accept 38.8 to 39.3)
Q14	MCQ	(A) its strength and hardness rise while its ductility falls
Q15	NAT	52.31 (accept 52.29 to 52.33)

Detailed Solutions

Q1 Answer: (B) Talkative

Concept: 'Taciturn' describes a person who says very little.

Reserved, silent and sullen are all close to it in meaning, not opposite.

The opposite is **talkative**, option (B).

Exam tip: For antonym questions, first strike out every option that is a synonym; what remains is usually short.

Q2 Answer: (B) 9:00 p.m.

Concept: A clock that gains at a steady rate is ahead by (gain per hour) × (hours elapsed).

From 8:00 a.m. to 8:00 p.m. is 12 hours, so the clock gains 12 × 5 = 60 minutes.

It therefore shows **9:00 p.m.**, option (B).

Exam tip: If the question gives the time shown by the faulty clock instead, the elapsed true time is (shown time)/(1 + gain fraction): here 65 clock minutes = 60 true minutes.

Q3 Answer: (A) (2, -1) and (-5, 3)

Concept: For a 2 × 2 matrix with rows (a, b), (c, d): A^{-1} = (1/det A) × [rows (d, −b), (−c, a)].

det A = 3 × 2 − 1 × 5 = 1.

So A^{-1} has rows **(2, −1) and (−5, 3)**, option (A).

Exam tip: Check by multiplying: (3, 1)·(2, −5) = 1 and (3, 1)·(−1, 3) = 0, giving the first row of I.

Q4 **Answer: 0.83**

Concept: Euler's method: $y_{n+1} = y_n + h\, f(x_n, y_n)$.

Step 1: $y_1 = 1 + 0.1 \times (0 - 1) = 0.9$ at $x = 0.1$.

Step 2: $y_2 = 0.9 + 0.1 \times (0.2 - 0.9) = \mathbf{0.83}$ at $x = 0.2$ (exact $2x - 2 + 3e^{-x} = 0.856$).

Exam tip: Euler's method uses only the slope at the start of each step, so its error per step is of order h^2 and the total error of order h.

Q5 **Answer: 50**

Concept: With uniform angular acceleration from rest, the angle turned = average speed × time.

Final speed = 600 rpm = 10 rev/s; average speed = 5 rev/s.

Revolutions = $5 \times 10 = \mathbf{50}$.

Exam tip: The rotational equations mirror the linear ones: $\theta = (\omega_0 + \omega)t/2$, just as $s = (u + v)t/2$.

Q6 **Answer: (B) 5/8**

Concept: $U = \Sigma\, P^2 L_i/(2A_iE)$; the load is fixed by the maximum stress in the thinnest part.

Both bars carry $P = \sigma A$ ($A = \pi d^2/4$). $U_A = \sigma^2 AL/(2E)$.

$U_B = (P^2/2E)[(L/2)/A + (L/2)/(4A)] = (\sigma^2 AL/2E)(1/2 + 1/8) = (5/8)U_A$; option (B).

Exam tip: Enlarging part of a bar lowers its energy-absorbing capacity for the same peak stress, which is why bolts for shock loads are turned down to the core diameter.

Q7 **Answer: 2.81 (accept 2.79 to 2.83)**

Concept: $\tau = VA'\bar{y}'/(Ib)$, which for a rectangle gives the parabola $\tau = (V/2I)(d^2/4 - y^2)$.

Average stress = V/(bd) = 72 000/28 800 = 2.5 MPa, so $\tau_{max} = 1.5 \times 2.5 = 3.75$ MPa at the neutral axis.

At y = 60 mm = d/4: $\tau = \tau_{max}(1 - 4y^2/d^2) = 3.75 \times (1 - 0.25) = \mathbf{2.81\ MPa}$.

Exam tip: Half-way from the neutral axis to the top fibre the shear stress is always 3/4 of its maximum in a rectangle.

Q8 **Answer: (D) 6 and 4**

Concept: The number of instantaneous centres of n links is $N = n(n - 1)/2$.

For $n = 4$, $N = 4 \times 3/2 = 6$.

Four of them (I_{12}, I_{23}, I_{34}, I_{14}) are at the pin joints and can be written down by inspection; the other two (I_{13}, I_{24}) are found with Kennedy's theorem: **6 and 4**, option (D).

Exam tip: Of the four at the joints, I_{12} and I_{14} (on the frame) are fixed; I_{23} and I_{34} move with the links.

Q9 **Answer: 15.49 (accept 15.40 to 15.58)**

Concept: Springs in series: $1/k_{eq} = 1/k_1 + 1/k_2$; $\omega_n = \sqrt{(k_{eq}/m)}$.

$k_{eq} = (2000 \times 3000)/(2000 + 3000) = 1200$ N/m.

$\omega_n = \sqrt{(1200/5)} = \sqrt{240} =$ **15.49 rad/s** (2.47 Hz).

Exam tip: In series the equivalent stiffness is always less than the smaller spring; in parallel the stiffnesses simply add.

Q10 **Answer: (A) 1.71**

Concept: Soderberg line: $\sigma_m/S_{yt} + \sigma_a/S_e = 1/N$.

$1/N = 100/300 + 50/200 = 0.3333 + 0.25 = 0.5833$.

$N = 1/0.5833 =$ **1.71**, option (A).

Exam tip: Soderberg (using S_{yt}) is the most conservative of the common criteria; Goodman uses S_{ut} and Gerber a parabola.

Q11 **Answer: 79.6 (accept 79.2 to 80.0)**

Concept: Hagen–Poiseuille: $\Delta p = 128\mu LQ/(\pi D^4)$, valid for laminar flow (Re below about 2000).

$V = Q/A = 0.001/(\pi \times 0.04^2/4) = 0.796$ m/s; $Re = \rho VD/\mu = 900 \times 0.796 \times 0.04/0.1 = 286$, so the flow is laminar.

$\Delta p = 128 \times 0.1 \times 50 \times 0.001/(\pi \times 0.04^4) = 0.64/(8.04 \times 10^{-6}) = 79\,600$ Pa = **79.6 kPa**.

Exam tip: At a fixed flow rate $\Delta p \propto 1/D^4$ in laminar flow, so a small error in diameter gives a large error in pressure drop.

Q12 **Answer: (C) 0.76 and 9.6 W**

Concept: For an insulated-tip fin, $\eta_f = \tanh(mL)/(mL)$ with $m = \sqrt{(hP/(kA))} = \sqrt{(4h/(kd))}$ for a pin.

$m = \sqrt{(4 \times 50/(200 \times 0.01))} = 10\ m^{-1}$, so $mL = 1$ and $\eta_f = \tanh 1 = 0.762$.

$q = \eta_f\, h\, (\pi dL)\, \theta_b = 0.762 \times 50 \times (\pi \times 0.01 \times 0.1) \times 80 =$ **9.6 W**; option (C).

Exam tip: Check with $q = \sqrt{(hPkA)}\, \theta_b \tanh(mL)$: $\sqrt{(hPkA)} = 0.157$ W/K, so $q = 0.157 \times 80 \times 0.762 = 9.6$ W.

Q13 **Answer: 39.0 (accept 38.8 to 39.3)**

Concept: Polytropic boundary work: $W_{by} = (p_1V_1 - p_2V_2)/(n - 1)$.

$p_2 = 100 \times 3^{1.3} = 417.1$ kPa; $p_2V_2 = 41.71$ kJ and $p_1V_1 = 30$ kJ.

$W_{by} = (30 - 41.71)/0.3 = -39.0$ kJ, so the work done **on** the air is **39.0 kJ**.

Exam tip: For compression the polytropic formula gives a negative value of work done by the gas; read the question for the sign it wants.

Q14 **Answer: (A) its strength and hardness rise while its ductility falls**

Concept: Cold work multiplies dislocations, which tangle and obstruct one another (strain hardening).

Yield strength, tensile strength and hardness therefore rise, while elongation (ductility) falls; conductivity drops slightly.

So **strength and hardness rise, ductility falls**, option (A).

Exam tip: Annealing (recovery, recrystallisation and grain growth) reverses these changes.

Q15 **Answer: 52.31 (accept 52.29 to 52.33)**

Concept: For an $\bar{x}$ chart built from ranges, UCL = $\bar{\bar{x}} + A_2\bar{R}$ and LCL = $\bar{\bar{x}} - A_2\bar{R}$.

$A_2\bar{R} = 0.577 \times 4.00 = 2.308$ mm.

UCL = 50.00 + 2.308 = **52.31 mm** (LCL = 47.69 mm).

Exam tip: $A_2\bar{R}$ is a shortcut for $3\sigma/\sqrt{n}$, with σ estimated as $\bar{R}/d_2$; A_2 falls as the sample size grows.

What next?

Day 85 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 86/90: Formula Sprint: Maths & GA. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.